

Mathematics A

Unit 1 • Chapter OL-T10 Classified

Cross-session • chapter-organised candidate

13 classified items • 39 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

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Contents

Unit 1 • Unit 1 • Chapter OL-T10 • 13 items • 39 marks

Chapter	Items	Marks	Starts
01. OL-T10 Surds	13	39	4

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Surds

Unit 1 • 39 marks • 13 item(s)

June 2025 | 4MA1-1H Q15 | 4 marks

Binomial surd denominator using a conjugate

June 2024 | 4MA1-1H Q17 | 3 marks

Binomial surd denominator using a conjugate

November 2023 | 4MA1-2H Q15 | 3 marks

Single-term surd denominator

June 2023 | 4MA1-2H Q23 | 4 marks

Binomial surd denominator using a conjugate

November 2024 | 4MA1-1H Q17 | 1 marks

Extract square factors from a surd

November 2024 | 4MA1-1H Q17 | 3 marks

Binomial surd denominator using a conjugate

June 2022 | 4MA1-1H Q16 | 3 marks

Add or subtract like surds

January 2022 | 4MA1-2H Q17 | 3 marks

Binomial surd denominator using a conjugate

January 2020 | 4MA1-1H Q17 | 3 marks

Expand binomials containing surds

January 2020 | 4MA1-2H Q20 | 3 marks

Binomial surd denominator using a conjugate

January 2021 | 4MA1-1H Q18 | 3 marks

Expand binomials containing surds

November 2021 | 4MA1-2H Q17 | 3 marks

Binomial surd denominator using a conjugate

May-Nov 2020 | 1H Q19 | 3 marks

Single-term surd denominator

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Working Unit 11

4MA1-1H Q15

UNIT 1
4 marks

15 $(\sqrt{3})^5 = k\sqrt{3}$ where k is an integer.

(a) Find the value of k

$$k = \dots\dots\dots$$

(1)

(b) Show that $\frac{21}{3 - \sqrt{2}}$ can be written in the form $c + \sqrt{d}$

where c and d are integers.

Show each stage of your working clearly.

(3)

WORKING SPACE

Question 12

4MA1-1H Q17

MODULAR UNIT 1

3 marks

- 17 Show that $\frac{1 + \sqrt{5}}{3 - \sqrt{5}}$ can be written in the form $a + \sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

WORKING SPACE

Question 27

4MA1-2H Q15 (b)

MODULAR UNIT 1

3 marks

- (b) Express $\frac{\sqrt{125} + \sqrt{80}}{\sqrt{3}}$ in the form $\sqrt{n}$ where n is an integer.

Show your working clearly.

.....
(3)

WORKING SPACE

Question 27

4MA1-2H Q23

MODULAR UNIT 1

4 marks

- 23 The diagram shows a cuboid with a square cross section.

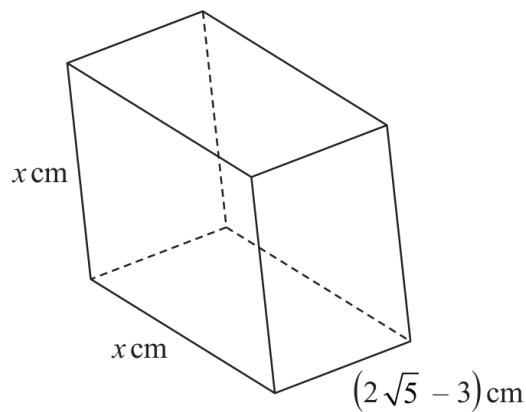


Diagram **NOT**
accurately drawn

The volume of the cuboid is $(13 + 6\sqrt{5})\text{cm}^3$

Without using a calculator, find the value of x

Give your answer in the form $a + \sqrt{b}$ where a and b are integers.

Show your working clearly.

$x = \dots\dots\dots$

Working space for Question 27

Continue your method

UNIT 1
4 marks

WORKING SPACE

Question 13

4MA1-1H Q17 M01

MODULAR UNIT 1

1 marks

17 (a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

.....
(1)

WORKING SPACE

Question 14

4MA1-1H Q17 M02

MODULAR UNIT 1

3 marks

- (b) Show that $\frac{5 - \sqrt{2}}{\sqrt{2} - 1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(3)

WORKING SPACE

Question 11

4MA1-1H Q16 Q16

MODULAR UNIT 1

3 marks

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$

Show your working clearly.

WORKING SPACE

Question 25

4MA1-2H Q17

MODULAR UNIT 1

3 marks

17 Show that $\frac{\sqrt{12}}{\sqrt{3} + 2}$

can be written in the form $a + \sqrt{b}$ where a and b are integers.

WORKING SPACE

17 (a) Show that $(6 + 2\sqrt{12})^2 = 12(7 + 4\sqrt{3})$

Show each stage of your working.

(3)

YOUR WORKING

20 The area of a rectangle is 18 cm^2

The length of the rectangle is $(\sqrt{7} + 1)\text{ cm}$.

Without using a calculator and showing each stage of your working,
find the width of the rectangle.

Give your answer in the form $a\sqrt{b} + c$ where a , b and c are integers.

..... cm

YOUR WORKING

18 Given that $(8 - \sqrt{x})(5 + \sqrt{x}) = y\sqrt{x} + 21$ where x is a prime number and y is an integer, find the value of x and the value of y . Show each stage of your working clearly.

$x =$

$y =$

YOUR WORKING

17 Express $\frac{8}{\sqrt{5}-1}$ in the form $\sqrt{a} + b$ where a and b are integers.

Show each stage of your working clearly.

YOUR WORKING

19 Without using a calculator, rationalise the denominator of $\frac{6}{3 - \sqrt{7}}$

Simplify your answer.

You must show each stage of your working.

(Total for Question 19 is 3 marks)

YOUR WORKING
